

2.44

Assuming unsigned numbers

a.00101010 

b.11011100 

c.01100101

Assuming 2's complement numbers

a.00101010 

b.11011100 

c.11100101

N.B: For the case of (b) 2's complement, shifting the bits caused the sign bit to flip hence changing it from a positive number to a negative number which led to an incorrect result.

2.51

a. 0 10000010 1001000000000000000000 

b. 1 01111110 1000000000000000000000 

c. 0 10000011 1010101000000000000000 

2.57

a. 1000000000000000000000100100111 

b. 10110010 00111001 00110101 

c. 00101001 01011100 

2.58

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ProbA.

-17.625 



Syndrome Check:

0

1

1 0

P0

Check:

Originally Stored 8-bit data: 1100 1101

Prob E.

$$2^p \geq 64 + p + 1$$

If $p=7$

$$2^7 \geq 64 + 7 + 1$$

Since $128 \geq 72$, 7 parity bits are sufficient.

Therefore, there are 7 parity bits to provide a 1-bit error correction, and 2-bit error detection for a 64-bit data word.

